

MAE HONG SON, Thailand

WMO: 483000

Lat: 19.301N

Lon: 97.976E

Elev: 283

StdP: 97.97

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	14.0	7.9	6.8	31.1	9.8	7.8	28.0	3.4	25.3	3.2	25.4	0.1	80	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	14.0	39.2	23.4	37.9	23.1	36.6	23.0	27.1	32.8	26.7	32.1	26.3	31.5	1.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.7	29.5	25.3	21.2	28.9	25.0	20.8	28.6	87.5	32.8	85.7	32.3	84.2	31.8	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
3.5	3.2	2.6	DB	11.0	41.0	1.5	1.4	9.9	42.0	9.0	42.8	8.2	43.6	7.1	44.7	
			WB	10.2	28.4	1.4	0.7	9.2	28.9	8.4	29.3	7.6	29.7	6.6	30.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	22.6	24.3	27.6	30.9	30.0	28.8	27.9	27.8	28.1	27.6	25.6	23.1
	DBStd	2.99	1.55	1.62	1.86	1.95	2.15	1.35	1.22	1.09	1.03	1.16	1.74	2.05
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6217	389	399	546	628	621	563	556	552	543	547	468	405
	CDD18.3	3176	132	166	288	378	362	313	298	293	293	288	218	147
	CDH23.3	31632	922	1617	3408	5082	4258	3263	2793	2678	2713	2487	1546	863
	CDH26.7	14152	299	789	1998	3114	2214	1276	906	841	957	957	559	241
(14) Wind	WSAvg	0.6	0.5	0.5	0.6	0.8	0.7	0.7	0.6	0.6	0.6	0.6	0.5	0.5
(15) Precipitation	PrecAvg	1289	9	6	20	57	175	184	225	246	201	116	41	9
	PrecMax	1679	74	45	130	138	362	290	414	367	363	257	172	61
	PrecMin	839	0	0	0	5	25	110	128	126	82	34	1	0
	PrecStd	204	17	11	34	43	73	54	67	65	74	55	38	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	35.8	38.6	41.1	40.6	36.1	34.6	33.8	34.2	34.2	33.1	31.9
		MCWB	21.0	20.3	21.6	23.8	24.5	25.7	26.1	26.1	25.9	25.4	24.2	22.5
	2%	DB	31.0	34.3	37.4	39.9	38.5	34.7	33.1	32.6	33.2	33.2	32.0	30.3
		MCWB	20.4	20.3	21.5	23.3	24.4	25.9	25.8	25.9	25.8	25.2	24.0	21.8
	5%	DB	29.8	33.1	36.5	38.8	36.7	33.3	32.0	31.8	32.3	32.4	31.1	29.2
		MCWB	20.1	20.0	21.3	23.2	24.9	25.7	25.6	25.7	25.7	25.1	23.6	21.4
	10%	DB	28.4	31.8	35.3	37.4	35.1	32.1	30.9	30.8	31.3	31.5	30.1	28.0
		MCWB	19.7	19.9	21.1	23.2	25.0	25.5	25.4	25.4	25.4	25.0	23.3	20.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	23.6	24.8	26.7	27.4	27.5	27.2	27.2	27.3	27.0	25.9	24.2
		MCDB	28.2	31.3	35.1	36.1	35.1	33.3	32.1	31.8	32.1	32.2	31.1	29.3
	2%	WB	22.3	22.5	23.9	25.8	26.7	26.8	26.6	26.6	26.7	26.3	25.1	23.4
		MCDB	27.3	30.3	33.6	35.2	34.1	32.4	31.3	31.0	31.4	31.4	29.8	28.0
	5%	WB	21.4	21.6	23.1	25.2	26.2	26.4	26.1	26.2	26.3	25.9	24.5	22.6
		MCDB	26.7	29.5	32.3	34.5	33.3	31.5	30.5	30.2	30.6	30.7	28.9	26.8
	10%	WB	20.7	20.8	22.4	24.7	25.7	25.9	25.7	25.7	25.8	25.4	24.0	22.0
		MCDB	26.1	28.7	31.4	33.5	32.2	30.6	29.6	29.4	29.9	29.8	28.2	26.0
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	17.4	17.0	14.0	10.1	7.6	6.7	6.7	7.6	9.0	10.4	11.8
		MCDBR	15.5	18.5	17.8	15.3	12.6	9.2	8.2	7.9	8.7	10.0	10.8	12.2
	5% WB	MCWBR	6.1	6.4	5.6	4.5	3.8	3.3	3.3	3.4	3.5	3.8	4.3	4.8
		MCWBR	13.0	16.4	15.6	13.3	10.5	8.0	7.1	7.2	7.9	9.0	9.7	10.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.398	0.471	0.667	0.656	0.518	0.452	0.438	0.450	0.469	0.491	0.420	0.401
		taud	2.312	2.106	1.638	1.729	2.133	2.359	2.398	2.350	2.276	2.172	2.326	2.332
	Ebn at Noon	Ebn at Noon	876	827	685	699	795	841	854	850	833	799	848	860
		Edh at Noon	121	157	260	239	158	125	120	127	135	145	119	116
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.77	5.64	5.86	6.03	5.34	4.51	4.10	4.12	4.51	4.60	4.67	4.42
	RadStd	RadStd	0.21	0.18	0.34	0.36	0.43	0.30	0.38	0.24	0.18	0.24	0.33	0.27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.16	N/A	N/A	+97	+98
(47) Regional (6 neighbors)	+0.29	+0.59	N/A	N/A	+0.15	+0.21	N/A	N/A	+147	+148

Nomenclature: See separate page